Means Separation and Data Presentation

AGRON 5130

YOUR NAME HERE

Introduction

The goal of this assignment is to give you hands-on practice with **means separation**, **linear contrasts**, and **presenting treatment means**.

You will work with an apple yield dataset to practice:

- fitting an ANOVA model
- comparing treatment means using **LSD** and **Tukey** comparisons
- constructing and testing **linear contrasts**
- presenting treatment means in a **figure with grouping letters**

Submit a **knitted PDF/HTML generated from your R Markdown file** to Canvas.

We can start by loading the necessary libraries

```r
library(ggplot2)
library(emmeans)
library(multcomp)
```

Question 1

1 point

Load the dataset `"data/apple.csv"` and display the first few rows of the dataset. This dataset contains apple yield data from an experiment that evaluated foliar treatments. The experiment was conducted as a randomized complete block design with four replications.

Question 2

2 points

Build a plot to visualize the relationship between `yield` and `foliar_trt`

Question 3

2 points

Fit an ANOVA model testing the effect of `foliar_trt` on apple yield.

Show the ANOVA table.

Question 4

2 points

Use the **emmeans** package to compute the estimated marginal means for `foliar_trt`.

Then conduct **pairwise comparisons using LSD inference**.

Show the output.

Question 5

3 points

Now conduct the same pairwise comparisons using **Tukey's HSD adjustment**.

Show the results and briefly explain how they differ from the LSD-style comparisons.

Question 6

3 points

Now test a **linear contrast** comparing

calcium plus additional ingredients

versus

calcium alone.

First examine the order of treatment levels in `foliar_trt`. Then compute the estimated marginal means using `emmeans`.

Construct the appropriate **contrast coefficients** and test the hypothesis using the `contrast()` function.

Briefly interpret the results.

Question 7

1 point

Use `cld()` to obtain **Tukey grouping letters** for the treatment means.

Show the output.

Question 8

1 point

Create a **plot using ggplot2** that shows:

- treatment means
- confidence intervals
- grouping letters above the bars

Your final figure should resemble the type of plot shown in the lecture.